PRODUCT USER MANUAL

For Mediterranean Sea Physical Reanalysis Product

MEDSEA_MULTIYEAR_PHY_006_004

Issue: 2.5

Contributors: Rafael Gomes de Menezes, Massimiliano Drudi, Alessandro Grandi, Emanuela Clementi

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00016

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	26.01.16	all	First version of document	R.Lecci, C. Fratianni, M. Drudi, A. Grandi	
1.1	01.04.16	all	Revision after AR2 following CMEMS remarks reported into CMEMS-MED-PUM-006-004-V2-V1.0_RevCoD.docx	R. Lecci, C. Fratianni	
1.2	18.01.17	all	General revision for CMEMS EIS V3.0	R. Lecci, C. Fratianni	
1.3	16.02.18	all	General revision for CMEMS EIS V4.1	R. Lecci	
1.4	21.01.19	all	Addition of analysis for 2017 year for CMEMS Q2/2019	R. Lecci, A. Grandi, E. Clementi, M. Drudi	
1.5	10.09.19	all	New template and general update of the document	R. Lecci, M. Drudi, E. Clementi	
2.0	10.09.20	all	New product nomenclature with new high resolution dataset. New template.	R. Lecci, M. Drudi, A. Grandi, E. Clementi, R. Escudier	C. Derval
2.1	15.01.21	all	Addition of interim datasets	R. Lecci, A. Grandi, E. Clementi	C. Derval
2.2	10.09.21	all	Updated text for Interim datasets re-initialization	R. Lecci, E. Clementi	C. Derval
2.3	29.11.22	all	Additional datasets	R. Lecci, A. Grandi, E. Clementi, M. Drudi	Copernicus Marine product managem ent
2.4	26.11.24	all	Addition of new datasets for air-sea fluxes	R. Lecci, A. Grandi, E. Clementi	Copernicus Marine product managem

					ent
2.5	28.05.25	all	Merge of Interim and Reanalysis datasets New dataset nomenclature Addition of daily, hourly and yearly Interim files	R. G. de Menezes, M. Drudi, A. Grandi, E. Clementi	Copernicus Marine product managem ent

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GLOSSARY AND ABBREVIATIONS

Analysis (Numerical)	A representation of the state of the ocean based on numerical modeling ingesting available observations in Near Real The operational production centre produces 4D time-space analysis data.
CF	Climate Forecast (convention for NetCDF)
CLS	Collecte Localisation Satellites
CMAP	CPC Merged Analysis of Precipitation
CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici
CMEMS	Copernicus Marine Environment Monitoring Service
CNR-ISAC	Istituto di Scienze dell'Atmosfera e del Clima
CTD	Conductivity Temperature Depth
DAC	Dynamic Atmospheric Correction
ECMWF	European Centre for Medium-Range Weather Forecasts
EOF	Empirical Orthogonal Function
FAQ	Frequently Asked Question
Forecast (Numerical)	A numerical forecast or prediction based on equations governing the motions and the forces affecting motion of fluids. The equations are based, or initialized, on specified ocean conditions at a certain place and time (NOAA Glossary).
MDT	Mean Dynamic Topography
Med/MED	Mediterranean

Meridional Velocity	South to North component of the horizontal velocity vector
MFC	Monitoring and Forecasting Centre
MFS	Mediterranean Forecasting System
NEMO	Nucleous for European Modelling of the Ocean
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
OA	Objective Analyses
OCEANVAR	Oceanographic variational data assimilation scheme developed at INGV/CMCC.
OGCM	Ocean General Circulation Model
OSI	Ocean and Sea Ice
PU	Production Unit
SL	Sea Level
SLA	Sea Level Anomaly
SSH	Sea Surface Height
SST	Sea Surface Temperature
TAC	Thematic Assembly Centre
XBT	eXpandable BathyThermograph
WW3	WaveWatch-III
Zonal Velocity	West to East component of the horizontal velocity vector
3DVAR	Three-Dimensional Variational

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the Copernicus Marine Mediterranean Sea MultiYear Physical Product **MEDSEA_MULTIYEAR_PHY_006_004** composed by:

- Reanalysis datasets including: 3D daily, monthly and yearly mean fields of Potential Temperature, Salinity, Zonal and Meridional Velocity; 2D daily, monthly and yearly mean fields of ocean mixed layer thickness and bottom potential temperature; 2D daily, hourly, monthly and yearly mean fields of Sea Surface Height; 2D daily and monthly air-sea fluxes fields, and hourly mean fields of Sea Surface Zonal and Meridional Velocity.
- Interim datasets including: 3D monthly mean fields of Potential Temperature, Salinity, Zonal and Meridional Velocity; 2D monthly mean fields of Ocean mixed layer thickness, bottom potential temperature and Sea Surface Height.

The Reanalysis timeseries is provided since 01/01/1987, it is extended once a year until 18 months before present (or at a higher frequency depending on the reprocessed observational data availability) and is available on the Copernicus Marine Data Store.

The Interim timeseries is produced using the Reanalysis modelling system but forced with NRT observational data, it is initialized with the last available day of Reanalysis and it is extended every month. Any time the Reanalysis timeseries is extended, the Interim timeseries is re-initialized and substituted on the Copernicus Marine Server.

MEDSEA_MULTIYEAR_PHY_006_004 product is organized in 25 datasets:

- 2 contain the hourly mean fields: 2D sea surface currents information and 2D sea surface height information.
    - **cmems_mod_med_phy-cur_my_4.2km_PT1H-m**
    - **cmems_mod_med_phy-ssh_my_4.2km_PT1H-m**
- 8 contain the 3D daily mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height,

bottom potential temperature, mixed layer thickness information and air-sea fluxes.

- **cmems_mod_med_phy-temp_my_4.2km_P1D-m**
- **cmems_mod_med_phy-cur_my_4.2km_P1D-m**
- **cmems_mod_med_phy-ssh_my_4.2km_P1D-m**
- **cmems_mod_med_phy-sal_my_4.2km_P1D-m**
- **cmems_mod_med_phy-mld_my_4.2km_P1D-m**
- **cmems_mod_med_phy-mflux_my_4.2km_P1D-m**
- **cmems_mod_med_phy-wflux_my_4.2km_P1D-m**
- **cmems_mod_med_phy-hflux_my_4.2km_P1D-m**

- 8 contain the monthly mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height bottom potential temperature, mixed layer thickness information and air-sea fluxes.
  - **cmems_mod_med_phy-temp_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-sal_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-cur_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-ssh_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-mld_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-mflux_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-wflux_my_4.2km_P1M-m**
  - **cmems_mod_med_phy-hflux_my_4.2km_P1M-m**
- 5 contain the yearly mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness information.
  - **cmems_mod_med_phy-tem_my_4.2km_P1Y-m**
  - **cmems_mod_med_phy-sal_my_4.2km_P1Y-m**
  - **cmems_mod_med_phy-cur_my_4.2km_P1Y-m**
  - **cmems_mod_med_phy-ssh_my_4.2km_P1Y-m**

- **cmems_mod_med_phy-mld_my_4.2km_P1Y-m**

- 1 contains the monthly climatological fields as mean and standard deviation of: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness information.
  - **cmems_mod_med_phy_my_4.2km-climatology_P1M-m**
- **1** contains the static fields for the system: coordinates, mean sea surface height, mask, and bathymetry: **cmems_mod_med_phy_my_4.2km_static**

The product is published on the Copernicus Marine Data Store after automatic and human quality controls. Product is available on-line and disseminated through the Copernicus Marine Information System. Files downloaded are in NetCDF format.

The reanalysis system is described in the Quality Information Document (QUID).

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

I.2 History of changes

Date	Description
Apr 2016	Change from NEMO version 3.2 to NEMO version 3.4
Sep 2019	New template and general revision
Sep 2020	New product nomenclature including: higher-resolution model grid, upgrade of the modelling system and forcing. New dataset. New template.
Jan 2021	Addition of interim datasets
Sep 2021	Updated text for Interim datasets re-initialization
Nov 2022	Addition of yearly and monthly climatology datasets
Nov 2024	Addition of air-sea fluxes datasets
May 2025	Merge of Interim and Reanalysis datasets New dataset nomenclature Addition of daily, hourly and yearly Interim files

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product name	MEDSEA_MULTIYEAR_PHY_006_004	
Geographical coverage	6°W → 36.29167°E; 30.1875°N → 45.97917°N	
Variables	Temperature Salinity Sea Surface Height Horizontal velocity (eastward and northward components) Ocean mixed layer thickness Sea floor potential temperature Wind Stress (eastward and northward components) Evaporation River Flux Precipitation Downward heat flux Net downward shortwave radiation flux Net downward longwave radiation flux Downward sensible heat flux Downward latent heat flux	
	Reanalysis	**Interim**
Update frequency	Yearly	Monthly
Available time series	Since 1987 up to the previous year	Since the last day of Reanalysis up to the previous month
Target delivery time	NA	20th of each month at 16UTC
Temporal resolution	cmems_mod_med_phy-cur_my_4.2km_PT1H-m, cmems_mod_med_phy-ssh_my_4.2km_PT1H-m:	

	hourly mean cmems_mod_med_phy-temp_my_4.2km_P1D-m, cmems_mod_med_phy-sal_my_4.2km_P1D-m, cmems_mod_med_phy-ssh_my_4.2km_P1D-m, cmems_mod_med_phy-mld_my_4.2km_P1D-m, cmems_mod_med_phy-cur_my_4.2km_P1D-m, cmems_mod_med_phy-mflux_my_4.2km_P1D-m, cmems_mod_med_phy-wflux_my_4.2km_P1D-m, cmems_mod_med_phy-hflux_my_4.2km_P1D-m: daily mean cmems_mod_med_phy-temp_my_4.2km_P1M-m, cmems_mod_med_phy-sal_my_4.2km_P1M-m, cmems_mod_med_phy-cur_my_4.2km_P1M-m, cmems_mod_med_phy-ssh_my_4.2km_P1M-m, cmems_mod_med_phy-mld_my_4.2km_P1M-m, cmems_mod_med_phy-mflux_my_4.2km_P1M-m, cmems_mod_med_phy-wflux_my_4.2km_P1M-m, cmems_mod_med_phy-hflux_my_4.2km_P1M-m: monthly mean cmems_mod_med_phy-tem_my_4.2km_P1Y-m, cmems_mod_med_phy-sal_my_4.2km_P1Y-m, cmems_mod_med_phy-cur_my_4.2km_P1Y-m, cmems_mod_med_phy-ssh_my_4.2km_P1Y-m, cmems_mod_med_phy-mld_my_4.2km_P1Y-m: yearly mean cmems_mod_med_phy_my_4.2km-climatology_P1M-m: monthly climatology
Horizontal resolution	1/24°
Number of vertical levels	141
Format	NetCDF CF1.6
Delivery mechanisms	MDS

Table 1: MED-MFC MultiYear products

The reanalysis time series is run twice a year using a daily assimilation cycle and assimilating best available reprocessed observation.

The Interim time series is run once a month using a daily assimilation cycle and assimilating available NRT observation. Any time the Reanalysis time series is extended, the Interim run is re-initialized and reproduced until one month before present.

II.2 Datasets details

MEDSEA_MULTIYEAR_PHY_006_004
Datasets: • **cmems_mod_med_phy-temp_my_4.2km_P1M-m** • **cmems_mod_med_phy-cur_my_4.2km_P1M-m** • **cmems_mod_med_phy-ssh_my_4.2km_P1M-m** • **cmems_mod_med_phy-mld_my_4.2km_P1M-m** • **cmems_mod_med_phy-sal_my_4.2km_P1M-m** • **cmems_mod_med_phy-mflux_my_4.2km_P1M-m** • **cmems_mod_med_phy-wflux_my_4.2km_P1M-m** • **cmems_mod_med_phy-hflux_my_4.2km_P1M-m** contain monthly mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness and air-sea fluxes • **cmems_mod_med_phy-temp_my_4.2km_P1D-m** • **cmems_mod_med_phy-cur_my_4.2km_P1D-m** • **cmems_mod_med_phy-ssh_my_4.2km_P1D-m** • **cmems_mod_med_phy-mld_my_4.2km_P1D-m** • **cmems_mod_med_phy-sal_my_4.2km_P1D-m** • **cmems_mod_med_phy-mflux_my_4.2km_P1D-m** • **cmems_mod_med_phy-wflux_my_4.2km_P1D-m** • **cmems_mod_med_phy-hflux_my_4.2km_P1D-m** contain daily mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness and air-sea fluxes • **cmems_mod_med_phy-cur_my_4.2km_PT1H-m**

• **cmems_mod_med_phy-ssh_my_4.2km_PT1H-m** contain hourly mean fields: 2D sea surface currents information and 2D sea surface height • **cmems_mod_med_phy-tem_my_4.2km_P1Y-m** • **cmems_mod_med_phy-sal_my_4.2km_P1Y-m** • **cmems_mod_med_phy-cur_my_4.2km_P1Y-m** • **cmems_mod_med_phy-ssh_my_4.2km_P1Y-m** • **cmems_mod_med_phy-mld_my_4.2km_P1Y-m** contain yearly mean fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness • **cmems_mod_med_phy_my_4.2km-climatology_P1M-m** contains the monthly climatology from 1993 to 2016 as mean and standard deviation fields: 3D potential temperature, salinity, and currents information from top to bottom and 2D sea surface height, bottom potential temperature, mixed layer thickness • **cmems_mod_med_phy_my_4.2km_static** contains the static fields for the system: coordinates, mean sea surface height, mask, and bathymetry
Variables name in the NetCDF file and Unit: Long_name & Standard_name
zos [m] Sea surface height sea_surface_height_above_geoid
zos_avg [m] Sea surface height sea_surface_height_above_geoid
zos_std [m] Sea surface height standard deviation sea_surface_height_above_geoid
uo [m/s] Eastward ocean current velocity eastward_sea_water_velocity
uo_avg [m/s] Eastward ocean current velocity eastward_sea_water_velocity
uo_std [m/s] Eastward ocean current velocity standard deviation eastward_sea_water_velocity
vo [m/s] Northward ocean current velocity northward_sea_water_velocity
vo_avg [m/s] Northward ocean current velocity

northward_sea_water_velocity
vo_std [m/s] Northward ocean current velocity standard deviation northward_sea_water_velocity
so [psu] Salinity sea_water_salinity
so_avg [psu] Salinity sea_water_salinity
so_std [psu] Salinity standard deviation sea_water_salinity
thetao [°C] Potential temperature sea_water_potential_temperature
thetao_avg [°C] Potential temperature sea_water_potential_temperature
thetao_std [°C] Potential temperature standard deviation sea_water_potential_temperature
bottomT [°C] Sea floor potential temperature sea_water_potential_temperature_at_sea_floor
bottomT_avg [°C] Sea floor potential temperature sea_water_potential_temperature_at_sea_floor
bottomT_std [°C] Sea floor potential temperature standard deviation sea_water_potential_temperature_at_sea_floor
mlotst [m] Ocean mixed layer thickness defined by density (as in de Boyer Montegut,2004) ocean_mixed_layer_thickness_defined_by_sigma_theta
mlotst_avg [m] Ocean mixed layer thickness defined by density (as in de Boyer Montegut,2004) ocean_mixed_layer_thickness_defined_by_sigma_theta
mlotst_std [m] Ocean mixed layer thickness defined by density (as in de Boyer Montegut,2004) ocean_mixed_layer_thickness_defined_by_sigma_theta
tauuo [N m-2] Wind Stress (u) surface_downward_x_stress
tauvo [N m-2]

Wind Stress (v) surface_downward_y_stress
evs [kg m-2 s-1] Evaporation surface_water_evaporation_flux
friver [kg m-2 s-1] River Flux water_flux_into_sea_water_from_rivers
pr [kg m-2 s-1] Precipitation precipitation_flux
hfds [W m-2] Downward heat flux surface_downward_heat_flux_in_sea_water
rsntds [W m-2] Net downward shortwave net_downward_shortwave_flux_at_sea_water_surface
rlds [W m-2] Net downward longwave radiation flux surface_net_downward_longwave_flux
hfss [W m-2] Downward sensible heat flux surface_downward_sensible_heat_flux
hfls [W m-2] Downward latent heat flux surface_downward_latent_heat_flux
e1t [m] Cell dimension along X axis
e2t [m] Cell dimension along Y axis
e3t [m] Cell dimension along Z axis cell_thickness
mask [1] Land-sea mask: 1 = sea ; 0 = land sea_binary_mask
deptho [m] Bathymetry sea_floor_depth_below_geoid
deptho_lev [1] Model level number at sea floor model_level_number_at_sea_floor
mdt [m] Mean dynamic topography

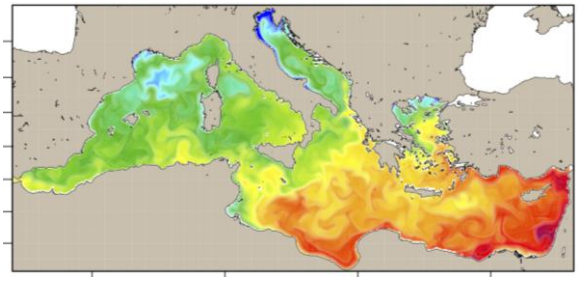
sea_surface_height_above_geoid

Table 2: list of the datasets and variable names and unit for the MEDSEA_MULTIYEAR_PHY_006_004 product

II.3 Additional Information on parameters

mlotst [m]	ocean_mixed_layer_thickness_defined_by_sigma_theta. It is the depth where the density increases of 0.01 kg.m-3 compared to density at 10 m depth
zos [m]	sea_surface_height_above_geoid. The geoid is a surface of constant geopotential (the mean ocean surface of the Earth if the ocean is at rest). The parameter "zos" is the difference between the actual sea surface height at any given time and place, and that which it would have if the ocean was at rest.

II.4 Production System Description

Domain **Resolution and grid** **Geographic coverage**	MEDSEA (6°W-36.29167°E; 30.1875°S – 45.97917°N) 1/24°; regular grid; 1016 x 380 x 141 This is a regional product covering the Mediterranean Sea. It is defined on a regular grid at 1/24 degree (approx. 4km) and 141 vertical levels. 
Model Version **Production unit** **Period of validation**	▪ NEMO v3.6 ▪ CMCC, Italy ▪ 01.01.1987–30.06.2021

Vertical grid	The product is delivered at the model's native grid, with up to 141 vertical levels: levels [m]: 1.02 3.17 5.46 7.92 10.54 13.32 16.27 19.40 22.71 26.20 29.89 33.77 37.85 42.15 46.65 51.38 56.33 61.52 66.95 72.62 78.55 84.74 91.20 97.93 104.94 112.25 119.86 127.77 136 144.55 153.43 162.66 172.24 182.18 192.48 203.17 214.25 225.72 237.61 249.92 262.65 275.83 289.46 303.56 318.13 333.19 348.75 364.82 381.42 398.54 416.22 434.46 453.27 472.67 492.67 513.29 534.53 556.41 578.94 602.15 626.03 650.62 675.91 701.93 728.69 756.2 784.47 813.53 843.39 874.06 905.55 937.89 971.08 1005.14 1040.08 1075.91 1112.66 1150.34 1188.95 1228.52 1269.05 1310.56 1353.07 1396.58 1441.11 1486.67 1533.27 1580.93 1629.65 1679.45 1730.33 1782.31 1835.41 1889.61 1944.95 2001.42 2059.03 2117.79 2177.71 2238.8 2301.06 2364.49 2429.11 2494.91 2561.9 2630.09 2699.47 2770.06 2841.84 2914.83 2989.02 3064.41 3141 3218.8 3297.79 3377.98 3459.37 3541.94 3625.7 3710.65 3796.77 3884.06 3972.52 4062.13 4152.9 4244.8 4337.85 4432.02 4527.3 4623.7 4721.19 4819.77 4919.43 5020.15 5121.93 5224.75 5328.59 5433.46 5539.33 5646.2 5754
Tides	N/A
Atmospheric forcing	Reanalysis: 1-h, 0.25° horizontal-resolution ERA5 reanalysis fields from the European Centre for Medium-Range Weather Forecasts (ECMWF) Interim: 1-h, 0.25° horizontal-resolution ERA5T reanalysis fields from the European Centre for Medium-Range Weather Forecasts (ECMWF)
Boundary forcing	The Mediterranean modeling system is forced in the Atlantic side by a Global Reanalysis run at CMCC: CGLORS-v5. The Dardanelles Strait inflow is instead parameterized as a river runoff.
Assimilation scheme	OceanVar (3DVAR)
Assimilated observations	In-situ vertical profiles of Temperature and Salinity from CTDs, XBTs, MBTs, bottles, ARGO floats; Sea Level Anomaly (SLA) from available satellites. Objective Analyses-Sea Surface Temperature (SST) fields are used to correct surface heat fluxes.

Initial conditions (spin-up process)	Reanalysis: Temperature and salinity monthly climatology (named SDN_V2aa) produced within the framework of SeaDataNet FP6 Project at 01/01/1985. Interim: Last day Reanalysis restart file
Bathymetry	GEBCO 30sec interpolated on the model grid
River run-off	Data from monthly climatological datasets

II.5 Processing information

II.5.1 Update Time

The Mediterranean Sea Physical reanalysis has been initialized by a temperature and salinity monthly climatology on the 1st of January 1985. The first two years are considered as period of model spin up.

The Mediterranean Sea Physical interim files are updated monthly on the 20th (addition of the previous month).

II.5.2 Temporal extent of reanalysis stored on delivery mechanism

The Reanalysis timeseries is available from 01/01/1987.

The Interim timeseries is available from the last day of Reanalysis.

II.5.3 Time averaging

For the yearly dataset, the fields are yearly averages over the calendar year (first to last day of the year). For the monthly dataset, the fields are monthly averages over the calendar month (first to last day of the month). For the daily dataset, the fields are daily averages over a day (00:00 to 24:00 UTC, centered at midday). For the hourly dataset, the fields are averaged in an hour (centered at HH:30).

III. FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV. FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

IV.1 Nomenclature of original files

The MEDSEA_MULTIYEAR_PHY_006_004 files nomenclature is based on the following pattern:

{valid date}_{freq flag}-{producer}--{parameter}-{config}-{region}-{bul date}_{product type}-sv{file version}.nc

where

- **valid date** YYYYMMDD is the validity day of the data in the file
- **freq flag** is the frequency of data values in the file (h = hourly, d = daily, m=monthly, y=yearly)
- **producer** is a short version of the Copernicus Marine Service production unit
- **config** identifies the producing system and configuration
- **region** is a three-letter code for the region
- **parameter** is a four-letter code for the parameter or parameter set from Standard BODC.
- **bul date** bYYYYMMDD is the bulletin date the product was produced
- **product type** is a two-letter code for the product type, for example fc for forecast, an for analysis, sm for hindcast, re for reanalysis and in for interim.
- **file version** is xx.yy where xx is the file version (01, 02 …) and yy is an incremental version number

Table 1 shows the nomenclature for the MEDSEA_MULTIYEAR_PHY_006_004 product.

Table 1 Description of the nomenclature for MEDSEA_MULTIYEAR_PHY_006_004

valid date	YYYYMMDD
freq flag	y (yearly)
	m (monthly)
	d (daily)
	h (hourly)
producer	CMCC
config	MFSe3r1 (reanalysis)
	MFSe3r1i (interim)
region	MED
parameter	TEMP
	PSAL
	ASLV
	RFVL
	AMXL
	MFLX
	WFLX
	HFLX
bulletin date	bYYYYYMMDD
product type	re (reanalysis) in (interim)
file version	01.00

Example for a file of Salinity for Reanalysis:

20180309_d-CMCC--PSAL-MFSe3r1-MED-b20200901_re-sv01.00.nc

This is the daily mean field of salinity centered at 12:00 UTC of the 9thMarch 2018, and the time coverage is from midnight (00:00 UTC) of the 9thMarch 2018 to 24:00 UTC of the 9thMarch 2018.

20180309_h-CMCC--RFVL-MFSe3r1-MED-b20200901_re-sv01.00.nc

This file contains the 24-hourly mean fields of currents, each one centered at 30' of every hour from midnight (00:00 UTC) of the 9thMarch 2018 to 24:00 UTC of the 9thMarch 2018.

20180301_m-CMCC--PSAL-MFSe3r1-MED-b20200901_re-sv01.00.nc

This is the monthly mean field of salinity for the month of March 2018. The mean is computed from midnight (00:00 UTC) of the 1st March 2018 to 24:00 UTC of the 31st March 2018.

20180101_y-CMCC--PSAL-MFSe3r1-MED-b20200901_re-sv01.00.nc

This is the yearly mean field of salinity for the year of 2018. The mean is computed from midnight (00:00 UTC) of the 1st January 2018 to 24:00 UTC of the 31st December 2018.

Example for a file of Salinity for Interim:

20200301_m-CMCC--PSAL-MFSe3r1i-MED-b20210101_in-sv01.00.nc

This is the monthly mean field of salinity for the month of March 2020. The mean is computed from midnight (00:00 UTC) of the 1st March 2020 to 24:00 UTC of the 31st March 2020.

IV.3 Other information: mean centre of Products, land mask value, missing value

The missing value for this product is: 1.e+20

Land mask is equal to "_FillValue" (see variable attribute on NetCDF file).

Real_Value = (Display_Value X scale_factor) + add_offset

The PHY reanalysis is using the NetCDF4 format without offset nor scale factors.

IV.4 File size

DATASET NAME	FILE NAME	DIMENSION [MB]
cmems_mod_med_phy-ssh_my_4.2km_P1D-m	{date1}_d-CMCC--ASLV-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_d-CMCC--ASLV-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.41
cmems_mod_med_phy-sal_my_4.2km_P1D-m	{date1}_d-CMCC--PSAL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_d-CMCC--PSAL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	20
cmems_mod_med_phy-temp_my_4.2km_P1D-m	{date1}_d-CMCC--TEMP-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_d-CMCC--TEMP-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	25.5
cmems_mod_med_phy-cur_my_4.2km_P1D-m	{date1}_d-CMCC--RFVL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_d-CMCC--RFVL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	73
cmems_mod_med_phy-mld_my_4.2km_P1D-m	{date1}_d-CMCC--AMXL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_d-CMCC--AMXL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.5
cmems_mod_med_phy-mflux_my_4.2km_P1D-m	{date1}_d-CMCC--MFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	0.82
cmems_mod_med_phy-wflux_my_4.2km_P1D-m	{date1}_d-CMCC--WFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	1.23
cmems_mod_med_phy-hflux_my_4.2km_P1D-m	{date1}_d-CMCC--HFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	2.05

cmems_mod_med_phy-cur_my_4.2km_PT1H-m	{date1}_h-CMCC--RFVL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_h-CMCC--RFVL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	23.5
cmems_mod_med_phy-ssh_my_4.2km_PT1H-m	{date1}_h-CMCC--ASLV-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_h-CMCC--ASLV-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	9.6
cmems_mod_med_phy-ssh_my_4.2km_P1M-m	{date1}_m-CMCC--ASLV-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_m-CMCC--ASLV-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.41
cmems_mod_med_phy-sal_my_4.2km_P1M-m	{date1}_m-CMCC--PSAL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_m-CMCC--PSAL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	20
cmems_mod_med_phy-temp_my_4.2km_P1M-m	{date1}_m-CMCC--TEMP-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_m-CMCC--TEMP-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	25.5
cmems_mod_med_phy-cur_my_4.2km_P1M-m	{date1}_m-CMCC--RFVL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_m-CMCC--RFVL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	73
cmems_mod_med_phy-mld_my_4.2km_P1M-m	{date1}_m-CMCC--AMXL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_m-CMCC--AMXL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.5
cmems_mod_med_phy-mflux_my_4.2km_P1M-m	{date1}_m-CMCC--MFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	0.82

cmems_mod_med_phy-wflux_my_4.2km_P1M-m	{date1}_m-CMCC--WFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	1.23
cmems_mod_med_phy-hflux_my_4.2km_P1M-m	{date1}_m-CMCC--HFLX-MFSe3r1-MED-b{date2}_re-sv01.00.nc	2.05
cmems_mod_med_phy-ssh_my_4.2km_P1Y-m	{date1}_y-CMCC--ASLV-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_y-CMCC--ASLV-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.41
cmems_mod_med_phy-sal_my_4.2km_P1Y-m	{date1}_y-CMCC--PSAL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_y-CMCC--PSAL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	20
cmems_mod_med_phy-tem_my_4.2km_P1Y-m	{date1}_y-CMCC--TEMP-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_y-CMCC--TEMP-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	25.5
cmems_mod_med_phy-cur_my_4.2km_P1Y-m	{date1}_y-CMCC--RFVL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_y-CMCC--RFVL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	73
cmems_mod_med_phy-mld_my_4.2km_P1Y-m	{date1}_y-CMCC--AMXL-MFSe3r1-MED-b{date2}_re-sv01.00.nc {date1}_y-CMCC--AMXL-MFSe3r1i-MED-b{date2}_in-sv01.00.nc	0.5
cmems_mod_med_phy_my_4.2km-climatology_P1M-m	19930101-20161231_m-CMCC--CLIM-MFSe3r1-MED-b{date2}_re-sv01.00.nc	3030
cmems_mod_med_phy_my_4.2km_static	MED-MFC_006_004_${field}.nc	4.2

V. REFERENCES

Quality Information Document:

https://documentation.marine.copernicus.eu/QUID/CMEMS-MED-QUID-006-004.pdf

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